Q1. What is an IP address?

1 mark

Ans: An IP address is a unique numerical identifier assigned to each device on a network.

Q2. Define latency in networking.

1 mark

Ans: Latency is the time delay between sending and receiving data.

Q3. What does CPU stand for?

1 mark

Ans: CPU stands for Central Processing Unit.

Q4. What is the full form of HTTP?

1 mark

Ans: HTTP stands for HyperText Transfer Protocol.

Q5. Differentiate between RAM and ROM (any 2 points).

2 marks

Ans: RAM is volatile memory and loses data when power is off, while ROM is non-volatile and retains data. RAM stores temporary data for active processing, whereas ROM stores permanent instructions like firmware. RAM is faster and easily modifiable, while ROM is slower and mostly read-only.

Q6. What is a firewall? Why is it used?

2 marks

Ans: A firewall is a network security system that monitors and controls incoming and outgoing traffic based on security rules. It is used to protect systems from unauthorized access, malware, and cyber attacks.

Q7. Explain the working of DNS in brief.

3 marks

Ans: The **Domain Name System (DNS)** is used to translate human-readable domain names into IP addresses.

Working:

1. When a user enters a domain (e.g., google.com), the request is sent to a DNS resolver.
2. The resolver checks its cache; if not found, it queries root, TLD, and authoritative DNS servers.
3. The correct IP address is returned to the client.
4. The browser then uses this IP to connect to the web server.

Q8. What is load balancing? Mention its importance.

3 marks

Ans: Load balancing is the process of distributing incoming network traffic across multiple servers to ensure no single server is overloaded.

Importance:

- Improves **system performance** by sharing workload
- Ensures **high availability** (if one server fails, others handle requests)
- Enhances **scalability** by allowing more servers to be added
- Reduces **response time** for users

Q9. Explain the OSI model and list its layers.

4 marks

Ans: The **OSI (Open Systems Interconnection) model** is a conceptual framework used to understand how data is transmitted over a network. It divides communication into seven layers, each with specific functions.

Layers:

1. **Physical Layer** – Transmits raw bits over a medium (cables, signals)
2. **Data Link Layer** – Handles error detection and frame transmission
3. **Network Layer** – Responsible for routing and IP addressing
4. **Transport Layer** – Ensures reliable data transfer (TCP/UDP)
5. **Session Layer** – Manages communication sessions
6. **Presentation Layer** – Handles data formatting, encryption
7. **Application Layer** – Interface for user applications (HTTP, FTP)

Q10. Explain how data is transmitted over the internet.

5 marks

Ans: Data transmission over the internet follows a structured process using networking protocols and devices:

1. **User Request:**
   The sender enters a URL in the browser. DNS translates it into an IP address.
2. **Data Packet Formation:**
   The data is divided into smaller units called **packets**. Each packet contains:
   - Source IP
   - Destination IP
   - Data
3. **Transmission via Network:**
   Packets travel through multiple **routers and switches** across different paths in the network.
4. **Protocol Handling:**
   Protocols like:
   - **TCP** ensure reliable delivery (error checking, ordering)

    - **IP** handles addressing and routing

5. **Reassembly:**
   At the receiver's end, packets are reassembled in the correct order to form the original data.

6. **Display:**
   The browser interprets and displays the data (webpage, video, etc.)

Key Components:

- Sender and Receiver
- DNS (Domain Name System)
- Routers & Switches
- Transmission Media (fiber, wireless)
- Protocols (TCP/IP)